

Reliability-Weighted Donor Prioritization (RWDP):

A Simulation-Based Evaluation of Commitment-Aware Matching for Emergency Blood Donation

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Abstract

Existing blood donation platforms in Pakistan function primarily as directories: they list donor contact information but do not reason about which donor is most likely to actually complete a donation. This paper presents Reliability-Weighted Donor Prioritization (RWDP), a matching algorithm implemented as part of ForiKhoon, an emergency blood donation platform, which ranks donors using a composite score combining blood-group compatibility, geographic proximity, real-time availability, and a longitudinal commitment score derived from confirmed donation outcomes. We evaluate RWDP against two baselines — pure random selection and an exact-match-only (V1) policy — using a Monte Carlo simulation (30 trials per scenario) across three donor-supply regimes: abundant (800 donors), moderate (300 donors), and scarce (150 donors). RWDP significantly improves fulfillment rate over both baselines under abundant and moderate supply ($p < 0.001$, $p = 0.008$) and consistently reduces no-show rates relative to random selection. Under severe scarcity, RWDP's prioritization advantage over random selection disappears ($p = 0.89$), while its advantage from compatibility-matrix matching persists ($p < 0.001$) — evidence that intelligent ranking and eligibility expansion are separable mechanisms with different value under different supply conditions. We also identify a fairness cost: RWDP concentrates load on top-ranked donors substantially more than random selection. We discuss this trade-off and outline a path toward a real-world pilot evaluation.

1. Introduction

Emergency blood donation in Pakistan remains largely dependent on informal, word-of-mouth coordination. Hospitals experiencing shortages have no structured way to reach willing donors quickly, and donors who are willing to help have no reliable channel through which they can be found. Several mobile directories exist that list registered donors by blood group and city, but none of the widely used tools reason about donor reliability — they surface a list and leave the burden of selection, contact, and follow-up entirely to the requester.

ForiKhoon was built to close this gap by treating donor selection as a ranking and matching problem rather than a lookup problem. This paper isolates and evaluates the core algorithmic contribution of that system: Reliability-Weighted Donor Prioritization (RWDP), a scoring model that combines medical compatibility, geography, availability, and a dynamically updated commitment score to decide which donors are contacted, and in what order, for a given request.

Our central research question is simple: does intelligent prioritization actually improve outcomes over naive selection, and if so, under what conditions does that advantage hold or break down? We answer this through controlled simulation rather than claims of intuition, and we report the conditions under which RWDP's advantage is statistically significant and the conditions under which it is not.

2. Related Work

Existing blood donation applications generally fall into two categories. The first is the directory model: donors register their blood group and city, and requesters browse or search the list manually. These systems provide reach but no prioritization — a rare-type donor and a currently unavailable donor are surfaced identically to a highly reliable, immediately available match. The second category is inventory-management software used internally by hospitals and blood banks, which tracks stock levels but is not designed to actively source donors from the public in real time.

The underlying scoring problem in RWDP is conceptually related to matching and prioritization problems studied in other on-demand allocation domains — for example, driver-rider matching in ride-hailing, where proximity and driver reliability jointly determine dispatch order. RWDP applies a similar composite-scoring philosophy to a domain with harder correctness constraints: blood-group compatibility is not a soft preference but a medical safety requirement, and rare blood types (O−, AB−) must be reserved rather than freely substituted across groups, which most general-purpose matching literature does not need to enforce.

3. Problem Formulation

Given a blood request specifying a blood group, urgency level, and hospital location, the system must select an ordered set of eligible donors to notify, such that: (1) every notified donor is medically compatible with the request, (2) rare blood types are never used as cross-type substitutes, (3) only donors who are currently available and outside their 90-day post-donation recovery window are considered, and (4) among eligible donors, those most likely to proximally respond and follow through are prioritized first. If no donor responds within a bounded time window, or a donor declines or fails to follow through after accepting, the system must escalate to a new candidate automatically rather than leaving the request unresolved.

4. The RWDP Algorithm

4.1 Scoring Model

Each eligible donor is scored as a weighted sum of four components:

- Exact blood-group match: +50 points
- Compatible (non-exact) match: +35 points
- Proximity: a gradient bonus of up to +30 points, fading linearly to 0 at 100km
- Availability: +20 points if the donor's availability toggle is currently on
- Commitment score bonus: $\text{commitment score} \times 0.5$, added on top of the above

Rare blood types (O−, AB−) are excluded from every other group's compatible-donor list at the eligibility stage, before scoring — they are only ever scored against requests for their own exact type, regardless of request urgency. This is a hard constraint, not a scoring penalty, since treating scarce-type reservation as a soft preference would allow it to be overridden under high urgency, defeating its purpose.

4.2 Commitment Score Dynamics

Every donor begins with a neutral commitment score, clamped between 0 and 100 throughout the donor's lifetime. The score updates only on confirmed outcomes, not on replies: a completed donation adds 10 points, a decline subtracts 5, and a no-show (accepting a match but never donating) subtracts 10 — penalized more heavily than a

decline because a no-show wastes the time lost to excluding other donors who were already passed over for that request. Accepting a match, on its own, does not change the score, since agreement is not evidence of reliability until it is followed through.

4.3 Radius-Based Search and Escalation

Eligible donors are searched at increasing radii — 10km, 25km, 50km, then 100km — using a bounding-box pre-filter followed by precise Haversine distance calculation on the resulting candidate set. The search stops at the first radius tier containing any qualifying donor, so nearby donors are always preferred over farther ones when both are available. The top three ranked donors from the winning tier are notified on initial request creation.

If a donor declines, or a hospital reports an accepted donor as a no-show, the system immediately re-runs the eligibility search — excluding every donor already tried for that request — and notifies a single replacement rather than waiting for a timeout. A separate background job runs every five minutes and escalates any request where every existing match is still pending after fifteen minutes of no response, notifying a fresh batch of up to three additional donors.

5. Experimental Setup

5.1 Simulation Design

We implemented a Monte Carlo simulation with synthetic donor and request populations, generated to reflect the blood-group distribution and rough geographic clustering assumptions used in the live system. Each of three donor-supply scenarios — abundant (800 donors), moderate (300 donors), and scarce (150 donors) — was run for 30 independent trials to account for stochastic variation in synthetic population generation and request arrival order.

5.2 Baselines

- Random selection: an eligible donor is chosen uniformly at random from the compatible pool, with no ranking.
- Exact-match-only (V1): the project's original pre-RWDP logic, which matches only exact blood-group donors and applies no proximity, availability, or commitment weighting.

5.3 Metrics

- Fulfillment rate: percentage of requests that ultimately resulted in a completed donation.
- No-show rate: percentage of accepted matches that did not result in a completed donation.
- Maximum donor load: the number of times the single most-contacted donor was reached within a trial, used as a proxy for load-distribution fairness.

6. Results

Table 1 summarizes fulfillment-rate outcomes across all three scenarios.

Scenario	Donors	RWDP Fulfillment	vs. Random	vs. Exact-Match-Only
Abundant	800	95.27%	+2.30pp (p<0.001)	+2.03pp (p<0.001)

Moderate	300	88.41%	+1.07pp (p=0.008)	+1.92pp (p<0.001)
Scarce	150	81.14%	+0.07pp (p=0.89, ns)	+2.84pp (p<0.001)

Table 1. Fulfillment rate by scenario, with significance relative to each baseline (pp = percentage points).

RWDP produced a statistically significant fulfillment-rate improvement over both baselines under abundant and moderate donor supply. It also produced a consistently lower no-show rate than random selection across every scenario — for example, 5.87% versus 11.83% under abundant supply — indicating that commitment-score weighting measurably improves donor reliability, not merely match speed or coverage.

Under severe donor scarcity (150 donors), RWDP's advantage over random selection was not statistically significant (p=0.89). Its advantage over the exact-match-only baseline remained significant (+2.84pp, p<0.001) in the same condition.

Table 2 summarizes the load-distribution fairness cost observed across scenarios.

Scenario	Max Donor Load — RWDP	Max Donor Load — Random
Abundant / Moderate / Scarce	~21–26 contacts	~8–19 contacts

Table 2. Maximum number of times the single most-contacted donor was reached within a trial.

7. Discussion

7.1 Why Ranking Loses Value Under Scarcity

The disappearance of RWDP's advantage over random selection at 150 donors is not a failure of the algorithm; it is a direct consequence of pool size. When the eligible candidate pool is small relative to the number of donors a request needs to try before finding a match, nearly every eligible donor ends up contacted regardless of the order in which they are approached. Prioritization changes who is contacted first, not who is eventually contacted — and when the pool is small enough that almost everyone is eventually contacted anyway, the order stops mattering much for the final outcome.

7.2 Two Separable Mechanisms

RWDP's benefit over the exact-match-only baseline persisted at every donor-supply level, including under scarcity, because that comparison isolates a different mechanism: compatibility-matrix matching expands who counts as an eligible donor in the first place, rather than reordering a fixed list. This distinction matters for interpreting the results — RWDP is really two contributions bundled together, an eligibility expansion (compatibility matching) and a ranking refinement (proximity, availability, commitment weighting) — and our results show these two contributions have different value profiles across supply conditions. Eligibility expansion helps regardless of scarcity; ranking refinement helps most when there is a meaningfully large pool to rank.

7.3 The Fairness Trade-off

RWDP consistently produces a higher maximum donor load than random selection across all three scenarios, since top-scored donors are repeatedly prioritized for new requests. This is an expected consequence of any reliability-weighted ranking system: donors who perform well are rewarded with more opportunities, which in a real deployment could translate into donor fatigue or burnout among the most reliable participants. We consider this an open fairness problem rather than a solved one, and discuss a mitigation direction below.

8. Limitations

- Synthetic data: donor and request populations were generated programmatically rather than sampled from real historical behavior. Real-world donor availability, response latency, and no-show correlation structures (for example, clustering around holidays or extreme weather) may differ from the simulation's assumptions, and could change the specific significance thresholds reported here.
- Single-region assumption: the simulation does not model cross-city variation in donor density, which is significant in Pakistan given the concentration of registered users in certain urban areas relative to smaller cities and rural regions.
- No behavioral calibration: response and no-show probabilities used in the simulation are modeled assumptions, not fitted to observed platform data, since the platform does not yet have a live user base large enough to calibrate against.
- Fairness metric is a proxy: maximum donor load approximates a fairness concern but does not capture donor-reported fatigue or willingness to continue participating, which would require real user feedback to measure directly.

9. Future Work

- Priority cooldown: introduce a temporary ranking penalty for donors who have been matched consecutively within a short window, to reduce load concentration without discarding reliability weighting entirely.
- Real-world pilot: deploy RWDP with a small set of partner hospitals and real donors, and replace synthetic behavioral assumptions with observed response and no-show data.
- Small-scale usability study: a System Usability Scale (SUS) evaluation with real donors and hospital staff, to validate that the workflow is usable in addition to being algorithmically sound.
- Learned scoring model: once sufficient real interaction data is available, evaluate whether a trained model (for example, logistic regression over the same feature set) outperforms the current fixed-weight formula.

10. Conclusion

RWDP demonstrates that intelligent, commitment-aware donor prioritization produces a measurable, statistically significant improvement in donation fulfillment and a meaningful reduction in no-show rates, under realistic donor-supply conditions. Its advantage is not unconditional: under severe scarcity, prioritization order stops mattering much because nearly the entire pool ends up contacted regardless, though eligibility-expanding compatibility matching continues to help. This bounded-advantage finding, together with the identified fairness trade-off in donor load distribution, gives a more honest and more useful picture of when algorithmic matching helps than a single aggregate accuracy number would — and sets a concrete agenda for the system's next iteration.